Paul Spencer

Define the conductor data

Phase Condcutor 250,000 AA:

$GMR_p := 0.0171 \qquad r_p := 0.4100 \qquad D_p := 0.567 \qquad i := \sqrt{-1}$

Neutral Conductor 4/0 6/1 ACSR:

$GMR_s := 0.00208 \qquad r_s := 14.87 \qquad D_s := 0.0641 \qquad D_{od} := 1.29$

$$R := \frac{D_{od} - D_s}{24} = 0.051 \qquad K := 13$$

$$r_{cn} := \frac{r_s}{K} = 1.144$$

$$GMR_{cn} := \sqrt[K]{GMR_s \cdot K \cdot R^{K-1}} = 0.049$$

Spacings between conductors and neutral:

$d_1 := 0 + 0\mathrm{i} \qquad d_2 := 0.5 + 0\mathrm{i} \qquad d_3 := 1 + 0\mathrm{i} \qquad d_4 := 0 + R \cdot i$

$d_5 := 0.5 + R \cdot i \qquad d_6 := 1 + R \cdot i$

$d_{11} := GMR_p \qquad d_{44} := GMR_{cn}$

$d_{22} := GMR_p \qquad d_{55} := GMR_{cn}$

$d_{33} := GMR_p \qquad d_{66} := GMR_{cn}$

$$D := \begin{bmatrix} d_{11} & |d_1 - d_2| & |d_1 - d_3| & |d_1 - d_4| & |d_1 - d_5| & |d_1 - d_6| \\ |d_2 - d_1| & d_{22} & |d_2 - d_3| & |d_2 - d_4| & |d_2 - d_5| & |d_2 - d_6| \\ |d_3 - d_1| & |d_3 - d_2| & d_{33} & |d_3 - d_4| & |d_3 - d_5| & |d_3 - d_6| \\ |d_4 - d_1| & |d_4 - d_2| & |d_4 - d_3| & d_{44} & |d_4 - d_5| & |d_4 - d_6| \\ |d_5 - d_1| & |d_5 - d_2| & |d_5 - d_3| & |d_5 - d_4| & d_{55} & |d_5 - d_6| \\ |d_6 - d_1| & |d_6 - d_2| & |d_6 - d_3| & |d_6 - d_4| & |d_6 - d_5| & d_{66} \end{bmatrix} = \begin{bmatrix} 0.017 & 0.5 & 1 & 0.051 & 0.503 & 1.001 \\ 0.5 & 0.017 & 0.5 & 0.503 & 0.051 & 0.503 \\ 1 & 0.5 & 0.017 & 1.001 & 0.503 & 0.051 \\ 0.051 & 0.503 & 1.001 & 0.049 & 0.5 & 1 \\ 0.503 & 0.051 & 0.503 & 0.5 & 0.049 & 0.5 \\ 1.001 & 0.503 & 0.051 & 1 & 0.5 & 0.049 \end{bmatrix}$$

$$Z_{p_ii} := r_p + \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{GMR_p}\right) + 7.93402\right)\right) = 0.505 + 1.456\mathrm{j}$$

$Z_{11} := Z_{p_ii} \qquad Z_{22} := Z_{p_ii} \qquad Z_{33} := Z_{p_ii}$

$$Z_{n_jj} := r_{cn} + \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{GMR_{cn}}\right) + 7.93402\right)\right) = 1.239 + 1.33\mathrm{j}$$

$Z_{44} := Z_{n_jj} \qquad Z_{55} := Z_{n_jj} \qquad Z_{66} := Z_{n_jj}$

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$$Z_{p_ij}(D_{ij}) := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{D_{ij}}\right) + 7.93402\right)\right)$$

$$Z_{12} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_1 - d_2|}\right) + 7.93402\right)\right) = 0.095 + 1.047\mathrm{j}$$

$$Z_{21} := Z_{12}$$

$$Z_{13} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_1 - d_3|}\right) + 7.93402\right)\right) = 0.095 + 0.963\mathrm{j}$$

$$Z_{31} := Z_{13}$$

$$Z_{14} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_1 - d_4|}\right) + 7.93402\right)\right) = 0.095 + 1.324\mathrm{j}$$

$$Z_{41} := Z_{14}$$

$$Z_{15} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_1 - d_5|}\right) + 7.93402\right)\right) = 0.095 + 1.046\mathrm{j}$$

$$Z_{51} := Z_{15}$$

$$Z_{16} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_1 - d_6|}\right) + 7.93402\right)\right) = 0.095 + 0.963\mathrm{j}$$

$$Z_{61} := Z_{16}$$

$$Z_{23} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_2 - d_3|}\right) + 7.93402\right)\right) = 0.095 + 1.047\mathrm{j}$$

$$Z_{32} := Z_{23}$$

$$Z_{24} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_2 - d_4|}\right) + 7.93402\right)\right) = 0.095 + 1.046\mathrm{j}$$

$$Z_{42} := Z_{24}$$

$$Z_{25} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_2 - d_5|}\right) + 7.93402\right)\right) = 0.095 + 1.324\mathrm{j}$$

$$Z_{52} := Z_{25}$$

$$Z_{26} := \left(0.09530 + 0.12134\mathrm{i} \cdot \left(\ln\left(\frac{1}{|d_2 - d_6|}\right) + 7.93402\right)\right) = 0.095 + 1.046\mathrm{j}$$

$$Z_{62} := Z_{26}$$

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$$Z_{34} := \left(0.09530 + 0.12134i \cdot \left(\ln\left(\frac{1}{|d_3 - d_4|}\right) + 7.93402\right)\right) = 0.095 + 0.963j$$

$$Z_{43} := Z_{34}$$

$$Z_{35} := \left(0.09530 + 0.12134i \cdot \left(\ln\left(\frac{1}{|d_3 - d_5|}\right) + 7.93402\right)\right) = 0.095 + 1.046j$$

$$Z_{53} := Z_{35}$$

$$Z_{36} := \left(0.09530 + 0.12134i \cdot \left(\ln\left(\frac{1}{|d_2 - d_6|}\right) + 7.93402\right)\right) = 0.095 + 1.046j$$

$$Z_{63} := Z_{36}$$

$$Z_{45} := \left(0.09530 + 0.12134i \cdot \left(\ln\left(\frac{1}{|d_4 - d_5|}\right) + 7.93402\right)\right) = 0.095 + 1.047j$$

$$Z_{54} := Z_{45}$$

$$Z_{46} := \left(0.09530 + 0.12134i \cdot \left(\ln\left(\frac{1}{|d_4 - d_6|}\right) + 7.93402\right)\right) = 0.095 + 0.963j$$

$$Z_{64} := Z_{46}$$

$$Z_{56} := \left(0.09530 + 0.12134i \cdot \left(\ln\left(\frac{1}{|d_5 - d_6|}\right) + 7.93402\right)\right) = 0.095 + 1.047j$$

$$Z_{65} := Z_{56}$$

$$Z_{prim} := \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} & Z_{14} & Z_{15} & Z_{16} \\ Z_{21} & Z_{22} & Z_{23} & Z_{24} & Z_{25} & Z_{26} \\ Z_{31} & Z_{32} & Z_{33} & Z_{34} & Z_{35} & Z_{36} \\ Z_{41} & Z_{42} & Z_{43} & Z_{44} & Z_{45} & Z_{46} \\ Z_{51} & Z_{52} & Z_{53} & Z_{54} & Z_{55} & Z_{56} \\ Z_{61} & Z_{62} & Z_{63} & Z_{64} & Z_{65} & Z_{66} \end{bmatrix} = \begin{bmatrix} 0.505 + 1.456j & 0.095 + 1.047j & 0.095 + 0.963j & 0.095 + 1.324j & 0.095 + 1. \\ 0.095 + 1.047j & 0.505 + 1.456j & 0.095 + 1.047j & 0.095 + 1.046j & 0.095 + 1. \\ 0.095 + 0.963j & 0.095 + 1.047j & 0.505 + 1.456j & 0.095 + 0.963j & 0.095 + 1. \\ 0.095 + 1.324j & 0.095 + 1.046j & 0.095 + 0.963j & 1.239 + 1.33j & 0.095 + 1. \\ 0.095 + 1.046j & 0.095 + 1.324j & 0.095 + 1.046j & 0.095 + 1.047j & 1.239 + 1. \\ 0.095 + 0.963j & 0.095 + 1.046j & 0.095 + 1.046j & 0.095 + 0.963j & 0.095 + 1. \end{bmatrix}$$

$$Z_{ij} := \text{submatrix}\left(Z_{prim}, 0, 2, 0, 2\right) = \begin{bmatrix} 0.505 + 1.456j & 0.095 + 1.047j & 0.095 + 0.963j \\ 0.095 + 1.047j & 0.505 + 1.456j & 0.095 + 1.047j \\ 0.095 + 0.963j & 0.095 + 1.047j & 0.505 + 1.456j \end{bmatrix}$$

Z_{in}:

$$Z_{nj} := Z_{in}$$

$$Z_{nn} := \text{submatrix}\left(Z_{prim}, 3, 5, 3, 5\right) = \begin{bmatrix} 1.239 + 1.33j & 0.095 + 1.047j & 0.095 + 0.963j \\ 0.095 + 1.047j & 1.239 + 1.33j & 0.095 + 1.047j \\ 0.095 + 0.963j & 0.095 + 1.047j & 1.239 + 1.33j \end{bmatrix}$$

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Phase Impedance Matrix

$$Z_{abc} := Z_{ij} - Z_{in} \cdot Z_{nn}^{-1} \cdot Z_{nj} = \begin{bmatrix} 0.798+0.447j & 0.319+0.033j & 0.289+0.055j \\ 0.319+0.033j & 0.789+0.405j & 0.311+0.111j \\ 0.289+0.055j & 0.311+0.111j & 0.691+0.617j \end{bmatrix}$$

Convert the phase impedance matrix to sequence impedance matrix

$a := 1\angle 120$ ***deg***

Define the sequence matrix, A_S

$$A_S := \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \qquad A_S = \begin{bmatrix} 1\angle 0^\circ & 1\angle 0^\circ & 1\angle 0^\circ \\ 1\angle 0^\circ & 1\angle -120^\circ & 1\angle 120^\circ \\ 1\angle 0^\circ & 1\angle 120^\circ & 1\angle -120^\circ \end{bmatrix}$$

Sequence Impedance Matrix

$$Z_{012} := A_S^{-1} \cdot Z_{abc} \cdot A_S = \begin{bmatrix} 1.372+0.623j & -0.051-0.081j & 0.084-0.007j \\ 0.084-0.007j & 0.453+0.423j & -0.024+0.012j \\ -0.051-0.081j & 0.073+0.034j & 0.453+0.423j \end{bmatrix}$$

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